UMA401 PROBABILITY AND STATISTICS

L	T	P	Cr
3	0	2	4

Course Objectives: This course shall make the students familiar with the concepts of Probability and Statistics useful in implementing various computer science models. One will also be able to associate distributions with real-life variables and make decisions based on statistical methods.

Introduction to Statistics and Data Analysis: Introduction to Statistical Inference, Samples, Populations and Experimental Design, Collection of Data, Measures of location and variability, Graphical representation of data.

Probability: Sample space, Events, Classical, relative frequency and axiomatic definitions of probability, addition rule and conditional probability, multiplication rule, total probability, Baye's Theorem.

Random Variables: Discrete, continuous and mixed random variables, probability mass, probability density and cumulative distribution functions, mathematical expectation, moments, probability and moment generating function, median and quantiles, Markov inequality, Chebyshev's inequality, Function of a random variable.

Special Distributions: Discrete uniform, binomial, geometric, negative binomial, Poisson, continuous uniform, exponential, gamma, normal, lognormal, inverse Gaussian, Cauchy, double exponential distributions, reliability of series and parallel systems.

Joint Distributions: Joint, marginal and conditional distributions, product moments, correlation and regression, independence of random variables, bivariate normal distribution.

Sampling Distributions: The Central Limit Theorem, distributions of the sample mean and the sample variance for a normal population, Chi-Square, t and F distributions.

Estimation: Unbiasedness, consistency, the method of moments and the method of maximum likelihood estimation, confidence intervals for parameters in one sample and two sample problems of normal populations, confidence intervals for proportions.

Testing of Hypotheses: Null and alternative hypotheses, the critical and acceptance regions, two types of error, power of the test, the most powerful test and Neyman-Pearson Fundamental Lemma, tests for one sample and two sample problems for normal populations, tests for proportions, Chi-square goodness of fit test and its applications.

Laboratory Work:
Implementation of statistical techniques using statistical packages viz. SPSS/R including evaluation of statistical parameters and data interpretation, regression analysis, covariance, hypothesis testing and analysis of variance.

Course Learning Outcomes (CLOs) / Course Objectives (COs):
After completion of this course, the students will be able to:

1. Analyze the data using different descriptive measures and present graphically.
2. Compute the probabilities of events along with an understanding of the random variables.
3. Comprehend the concept of statistical distributions, their properties and relevance to real-life data.
4. Understand the estimation of mean and variance and their respective hypothesis tests.

Text Books:

1. Probability & Statistics for Engineers & Scientists by R.E. Walpole, R.H. Myers, S.L. Myers & Keying Ye, Prentice Hall, (2016), 9^{th} edition.
2. An Introduction to Probability and Statistics by V.K. Rohatgi & A.K. Md. E. Saleh, Wiley, (2008), 2^{nd} edition

Reference Books:

1. Miller and Freund's – Probability and Statistics for Engineers by R. A. Johnson, Person Education, (2017), 9^{th} edition.
2. Introduction to Probability and Statistics for Engineers and Scientists by S.M. Ross, Elsevier, (2014), 4^{th} edition.